

Randomized Block Design Analysis of Wheat Yield

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1. Objective

The objective of this study is to determine whether wheat yield differs:

1. Between geographical areas (treatments),
2. Between years (blocks),
3. And whether there is an interaction between area and year.

The experiment follows a Randomized Block Design (RBD) with replication, where:

- Treatments = 3 Areas
 - Blocks = 3 Years
 - Replications per cell = 2 plots
 - Response = Yield (kg)
-

2. Statistical Model

The model used is:

$$Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \epsilon_{ijk}$$

where:

- μ = overall mean
 - α_i = area effect
 - β_j = year effect
 - $(\alpha\beta)_{ij}$ = interaction
 - $\epsilon_{ijk} \sim N(0, \sigma^2)$
-

3. Hypotheses Tested (5% level)

(i) Area effect

$$H_0 : \alpha_1 = \alpha_2 = \alpha_3$$

H_1 : At least one α_i is different

(ii) Year effect

$$H_0 : \beta_1 = \beta_2 = \beta_3$$

H_1 : At least one β_j is different

(iii) Interaction

$$H_0 : (\alpha\beta)_{ij} = 0$$

$$H_1 : (\alpha\beta)_{ij} \neq 0$$

4. Data Entry

```
area <- rep(c("A1", "A2", "A3"), each = 2, times = 6)
year <- rep(c("Y1", "Y2", "Y3"), each = 12)
repno <- rep(1:2, times = 18)

yield <- c(
  14.20, 17.45, 13.6, 19.22, 16.96, 16.3,
  14.40, 16.00, 16.28, 25.4, 16.62, 14.3,
  14.98, 15.9, 17.02, 14.3, 12.16, 14.34,
  15.80, 17.58, 13.96, 13.1, 16.89, 13.46,
  14.34, 12.14, 12.0, 14.74, 17.83, 16.22,
  11.96, 15.24, 14.5, 12.86, 18.03, 15.12)

wheat <- data.frame(area, year, repno, yield)

head(wheat)
```

```
##   area year repno yield
## 1   A1   Y1     1 14.20
## 2   A1   Y1     2 17.45
## 3   A2   Y1     1 13.60
## 4   A2   Y1     2 19.22
## 5   A3   Y1     1 16.96
## 6   A3   Y1     2 16.30
```

5. Fit the RBD Model

```
model <- aov(yield ~ area * year, data = wheat)
summary(model)
```

```
##              Df Sum Sq Mean Sq F value Pr(>F)
## area          2   3.29   1.643   0.338 0.7161
## year          2  31.52  15.758   3.242 0.0547 .
## area:year      4  56.08  14.021   2.885 0.0413 *
## Residuals    27 131.23   4.860
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

6. Results and Interpretation

From the ANOVA table: Area: $p = 0.7161 > 0.05$ (Not significant)

Year: $p = 0.0547 > 0.05$ (Not significant)

Interaction (Area $\times$ Year): $p = 0.0413 < 0.05$ (Significant)

Since the interaction is significant, the effect of the geographical area on yield depends on which year is being observed.

7. ConclusionAt the 5% level of significance:

We accept H_0 for the Area effect; the geographical areas are not significantly different on average.

We accept H_0 for the Year effect; the blocks (years) are not significantly different on average.

We reject H_0 for the Interaction; there is a significant interaction between area and year.

Final Note: Because the interaction is significant, we should focus on the specific combinations of Area and Year rather than looking at the average performance of an Area across all years.